

SOC138: Malicious Macro-Enabled Excel File

Incident Investigation Report: C2 Communication Confirmed

Platform	LetsDefend (Malware Playbook, Security Analyst)
Analyst	Asha Latha Panuganti
Verdict	True Positive
Action taken	Host contained via EDR
Playbook accuracy	100%

1. Summary

A macro-enabled Excel file (**ORDER SHEET & SPEC.xlsm**) was delivered to host **Sofia (172.16.17.56)** and allowed by the security control. Threat intelligence showed the file exploits **CVE-2017-11882** and downloads a second-stage executable from **multiwaretecnologia.com.br**. Firewall logs confirmed the host connected to the C2 IP **177.53.143.89:443** at the time of execution. The malware was not quarantined, so the host was contained.

Key analytical step: the C2 domain's *current* DNS record did not match the IP in the logs. Historical (passive) DNS showed the domain resolved to 177.53.143.89 during the incident period, which linked the network activity to the malware.

2. Alert Details

Field	Value
Rule	SOC138 - Detected Suspicious Xls File
Event ID	77
Event time	2021-03-13 20:20:58 (+03:00) = 17:20:58 UTC
Severity	Medium
Alert type	Malware
Host	Sofia
Source IP	172.16.17.56
File name	ORDER SHEET & SPEC.xlsm
File size	2.66 MB
MD5	7ccf88c0bbe3b29bf19d877c4596a8d4
Device action	Allowed

Medium	2021-03-13T20:20:58+03:00	SOC138 - Detected Suspicious Xls File
EventID :	77	
Event Time :	2021-03-13T20:20:58+03:00	
Rule :	SOC138 - Detected Suspicious Xls File	
Role :	Security Analyst	
Alert Type :	Malware	
Difficulty :	Easy	
MITRE ATT&CK :	T1112	
File Hash :	7ccf88c0bbe3b29bf19d877c4596a8d4	
File Name :	ORDER SHEET & SPEC.xlsm	
File Size :	2.66 Mb	
Device Action :	Allowed	
Source Address :	172.16.17.56	
Source Hostname :	Sofia	

Figure 1: SOC138 alert details in LetsDefend

3. Investigation Steps

3.1 Quarantine check

The alert's **Device Action = Allowed**, meaning the security control did not block or remove the file. **Result: not quarantined**, so the threat was potentially active on the endpoint.

3.2 File hash analysis (VirusTotal)

Detection: 46/64 vendors flagged the file as malicious; community score -10. Relevant behaviour tags:

- **macros, auto-open, macro-run-file:** code executes automatically when the file is opened
- **cve-2017-11882, exploit:** exploits the Microsoft Office Equation Editor vulnerability
- **exe-pattern, write-file, executes-dropped-file, run-dll:** drops and runs an executable (downloader behaviour)
- **detect-debug-environment, long-sleeps:** sandbox and analysis evasion

Verdict: Malicious. A legitimate order spreadsheet has no reason to exploit a CVE or execute dropped files.

46 / 64
Community Score -10

46/64 security vendors flagged this file as malicious

7bcd31bd41686c32663c7cabf42b18c50399e3b3b4533fc2f002d9f2e058813
ORDER SHEET & SPEC.xlsm

Size 2.66 MB | Last Analysis Date 4 days ago | XLSX

xlsm clipboard checks-user-input open-file cve-2017-11882 exploit run-file macro-run-file auto-open exe-pattern long-sleeps calls-wmi detect-debug-environment
executes-dropped-file write-file run-dll macros

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 30+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Basic properties

MD5	7ccf88c0bbe3b29bf19d877c4596a8d4
SHA-1	23f0506d857d38c3cd5354b80afc725b5f034744
SHA-256	7bcd31bd41686c32663c7cabf42b18c50399e3b3b4533fc2f002d9f2e058813
Vhash	b513aac5dd6acbd349d532900f637d7
SSDEEP	1536:HHh3S1cLkPROxYvovIZCMV2Zx0nlcjELcE3E:0cCOxtYIEbsX0n98E
TLSH	T159D54410F2177A42C5DBE23802E227A4776443B48139B897BF61B389C923F537D9E1A
File type	Office Open XML Spreadsheet document msoffice spreadsheet excel xlsm
Magic	Microsoft Excel 2007+
TrID	Excel Microsoft Office Open XML Format document (with Macro) (45.9%) Excel Microsoft Office Open XML Format document (27.1%) Open Packaging Conventions container (13.9%) ...
File size	2.66 MB (2793487 bytes)

History

Figure 2: VirusTotal detection (46/64) and behaviour tags

3.3 Identifying the C2

VirusTotal Relations tab, Contacted URLs:

URL	Detections	Assessment
https://multiwaretecnologia.com.br/js/Podaliri4.exe	8/90	Malicious: executable hosted in a /js/ path on an unrelated site, typical of a compromised payload host
http://ocsp.pki.goog/...	0/90	Benign: Google certificate revocation check

Other contacted domains (*.in-addr.arpa, akamai.net) are normal sandbox noise (reverse DNS lookups and CDN traffic) and were excluded.



The screenshot shows the VirusTotal interface. Under 'Contacted URLs (2)', there are two entries: one from 2026-06-04 with 0/90 detections and a status of 200, and another from 2026-09-22 with 8/90 detections and a status of 404. The 404 status is highlighted in red. Under 'Contacted Domains (23)', there are two entries for *.in-addr.arpa domains, both with 0/89 detections.

Scanned	Detections	Status	URL
2026-06-04	0 / 90	200	http://ocsp.pki.goog/gsr2/ME4wTDBKMEgwRjAJBgUrDgMCGGUABBTgXlsxbvr2IBkPpolEVRE6gHICnAQUm+IHV2ccHsBqBt5ZtJot39wZhi4CDQHjtJqhYqpgSVpULg=
2026-09-22	8 / 90	404	https://multiwaretecnologia.com.br/js/Podaliri4.exe

Domain	Detections	Created	Registrar
12.179.89.13.in-addr.arpa	0 / 89	-	-
20.173.189.20.in-addr.arpa	0 / 89	-	-

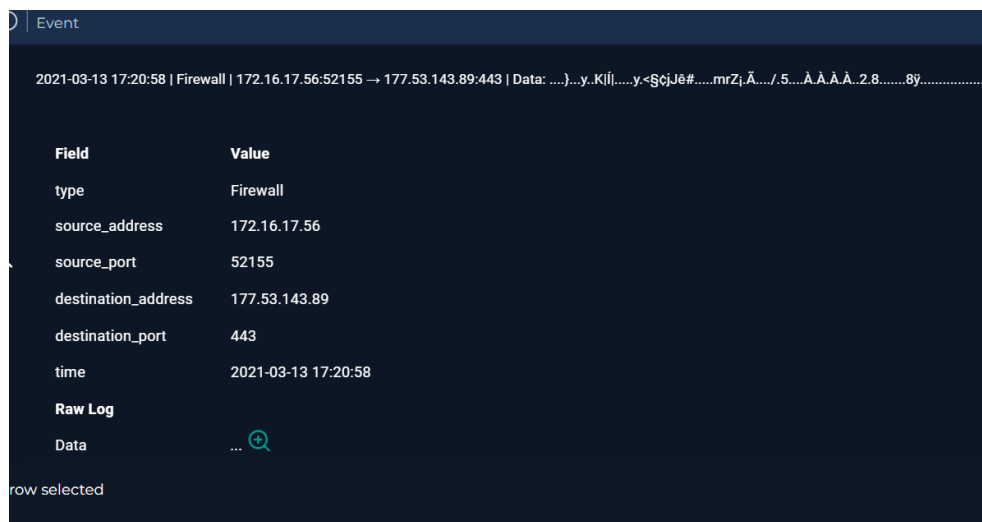
Figure 3: Contacted URLs, showing the malicious payload URL

3.4 Log correlation

Searched Log Management for traffic from 172.16.17.56 around the alert time:

Time (UTC)	Type	Source	Destination
2021-03-13 17:20:10	Firewall	172.16.17.56:52155	177.53.143.89:443
2021-03-13 17:20:58	Firewall	172.16.17.56:52155	177.53.143.89:443

The alert time (20:20:58 +03:00) converts to **17:20:58 UTC**, matching these connections. An unrelated proxy log from October 2020 (stylefix.co) was reviewed and excluded because it falls outside the incident window.



The screenshot shows a Firewall log entry for the time 2021-03-13 17:20:58. The log details a connection from source address 172.16.17.56, source port 52155, to destination address 177.53.143.89, destination port 443. The time is 2021-03-13 17:20:58. There is a 'Raw Log' section with a 'Data' field and a magnifying glass icon.

Field	Value
type	Firewall
source_address	172.16.17.56
source_port	52155
destination_address	177.53.143.89
destination_port	443
time	2021-03-13 17:20:58

Raw Log

Data ... 🔍

Figure 4: Firewall log showing Sofia connecting to 177.53.143.89:443

3.5 Linking the IP to the C2 domain (Passive DNS pivot)

Problem: the domain's current A record resolves to 177.11.52.83, not the 177.53.143.89 seen in the logs.

Approach: instead of assuming the IP was unrelated, I checked historical DNS resolution.

Date resolved	IP
2019-11-22	177.11.48.112
2020-07-23	177.53.143.89
2025-04-15	177.11.52.83

- The domain resolved to **177.53.143.89** from July 2020, with no change recorded until April 2025, which covers the incident date (March 2021).
- Supporting evidence: the domain's SPF (TXT) record authorises 177.53.143.0/24, and several subdomains also resolved to 177.53.143.89.

Result: C2 accessed. Sofia communicated with the C2 infrastructure at the time of execution.

Passive DNS Replication (3) ⓘ			
Date resolved	Detections	Resolver	IP
2025-04-15	0 / 89	VirusTotal	177.11.52.83
2020-07-23	0 / 89	VirusTotal	177.53.143.89
2019-11-22	1 / 89	VirusTotal	177.11.48.112

Subdomains (10) ⓘ		
Subdomain	Detections	IPs
multiwaretecnologia.com.br	6 / 89	177.11.52.83 177.53.143.89 177.11.48.112
cpcontacts.multiwaretecnologia.com.br	0 / 89	177.53.143.89
cpcalendars.multiwaretecnologia.com.br	0 / 89	177.53.143.89
autodiscover.multiwaretecnologia.com.br	0 / 89	177.11.48.112
webmail.multiwaretecnologia.com.br	0 / 89	177.53.143.89 177.11.48.112
mail.multiwaretecnologia.com.br	0 / 89	177.53.143.89 177.11.48.112
topsecreto.multiwaretecnologia.com.br	0 / 89	177.11.52.83 177.53.143.89 177.11.48.112

Figure 5: Passive DNS replication for multiwaretecnologia.com.br

3.6 Endpoint review and containment

- The EDR process list showed **POwersheLL.exe**. The unusual mixed-case naming is a known technique for evading simple string-based detection rules. No command-line data was available to confirm its activity.
- Host **Sofia** was **contained** via EDR to prevent further C2 communication and lateral movement.

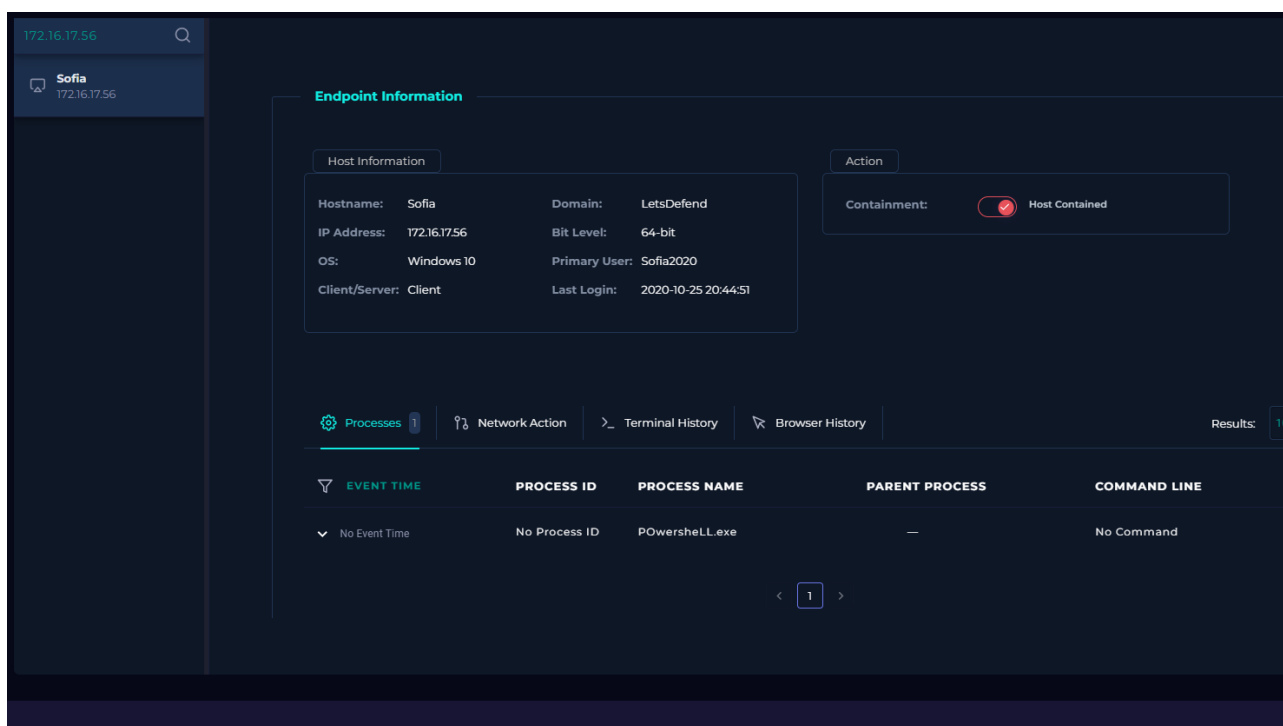


Figure 6: Sofia contained in EDR

4. Timeline (UTC)

Time	Event
2021-03-13 ~17:20	ORDER SHEET & SPEC.xlsm opened on Sofia; macro executes
17:20:10	First connection from Sofia to C2 177.53.143.89:443
17:20:58	Second connection to C2; SOC138 alert triggered
Investigation	Analyst investigation; host contained

5. Indicators of Compromise (IOCs)

Type	Value
MD5	7ccf88c0bbe3b29bf19d877c4596a8d4
SHA-256	7bcd31bd41686c32663c7cabf42b18c50399e3b3b4533fc2ff002d9f2e058813
File name	ORDER SHEET & SPEC.xlsm
URL	https://multiwaretecnologia.com.br/js/Podaliri4.exe
Domain	multiwaretecnologia.com.br
IP (C2, 2021)	177.53.143.89

6. MITRE ATT&CK Mapping

Technique	ID	Evidence
User Execution: Malicious File	T1204.002	User opened the .xlsm file
Exploitation for Client Execution	T1203	CVE-2017-11882 tag on VirusTotal
Ingress Tool Transfer	T1105	Download of Podaliri4.exe from C2
Modify Registry	T1112	Technique assigned by the detection rule
Command and Scripting Interpreter: PowerShell	T1059.001	Possible: POWERSHELL.exe observed on endpoint; not confirmed

7. Verdict and Response

Verdict: True Positive. Malicious file executed and communicated with C2.

Actions taken:

- Host Sofia contained via EDR
- IOCs documented and added to the case

Recommended next steps:

- Block the C2 domain and IP at the firewall/proxy
- Search all endpoints for the file hash and for connections to 177.53.143.89
- Identify the delivery vector (e.g. check Email Security for the original attachment and other recipients)
- Reset user Sofia2020's credentials in case of credential theft
- Reimage the host before returning it to the network
- Ensure Office patches for CVE-2017-11882 are applied across the estate

8. Lessons Learned

- **Current DNS is not historical DNS.** Infrastructure changes over time. Passive DNS was essential to connect a 2021 log entry to the malicious domain.
- **Always normalise time zones** before correlating alert times with log timestamps.
- **Filter sandbox noise** (OCSP, reverse DNS, CDNs) to focus on the indicators that matter.

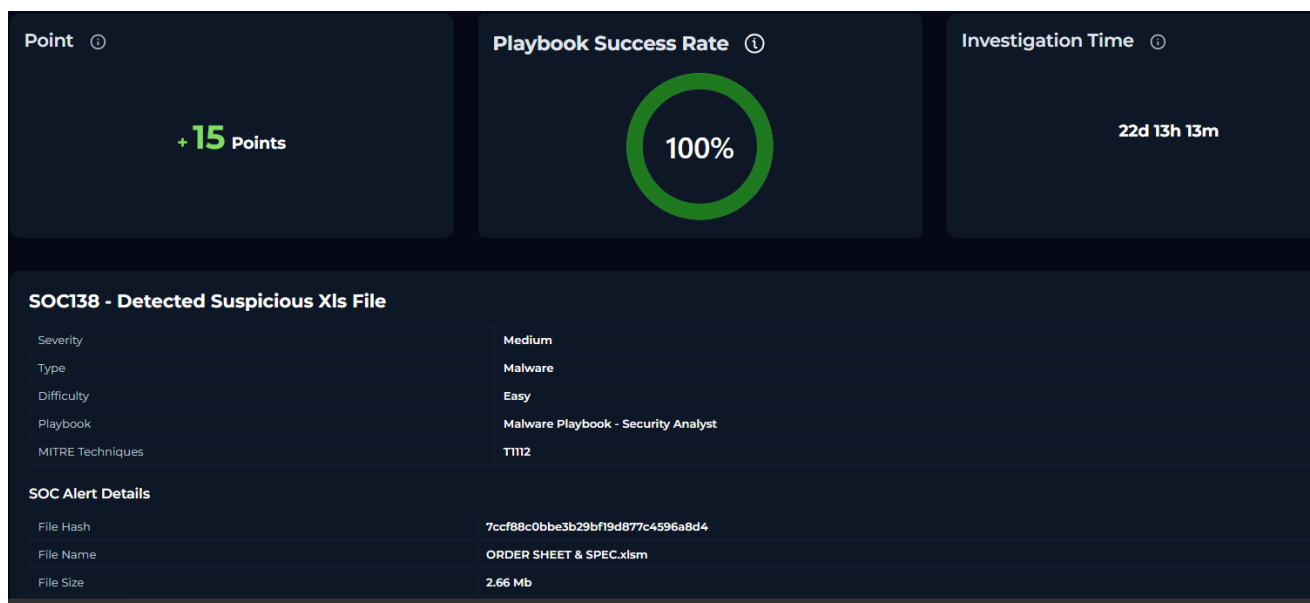


Figure 7: Playbook result, 100% success rate

Tools used: LetsDefend SIEM, Log Management, EDR; VirusTotal (detection, relations, passive DNS)